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14 November 1975

Superseding
MIL-T-81038C(AS)
16 March 1971

MILITARY SPECIFICATION

TRACTOR, AIRCRAFT TOWING, GASOLINE 7500 POUNDS DRAWBAR PULL, TYPE TA-75

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers a 4 x 2 type gasoline engine powered tractor intended for use as a towing vehicle for aircraft with gross weights up to 75,000 pounds.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

GG-G-76	Gages, Pressure and Vacuum, Dial Indicating, (For Air, Steam, Oil, Water, Ammonia)
TT-L-54	Lacquer, Spraying, Acid-Resistant
VV-G-001690	Gasoline, Automotive, Low Leaded or Unleaded

FSC 2420

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SPECIFICATIONS (Cont'd)

Military

MIL-P-116	Preservation, Methods of
MIL-P-514	Plates; Identification, Instruction and Marking, Blank
MIL-F-3541	Fittings, Lubrication
MIL-T-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-M-8090	Mobility, Towed Aerospace Ground Equipment, General Requirements For
MIL-A-8421	Air Transportability Requirements, General Specification For
MIL-T-8504	Tubing Steel, Corrosion Resistant (304) Aerospace Vehicle Hydraulic Systems, Annealed, Seamless and Welded
MIL-S-8512	Support Equipment, Aeronautical, Special, General Specification for the Design of
MIL-S-8516	Sealing Compound, Synthetic Rubber, Electric Connectors and Electrical Systems
MIL-B-11188	Batteries, Storage, Lead-Acid (Starting, Lighting, and Ignition, Automotive Type)

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SPECIFICATIONS (Cont'd)

Military

MIL-T-12459	Tire, Pneumatic; for Military Ground Vehicles
MIL-D-17951	Deck Covering, Lightweight, Non Slip Silicon Carbide Particle Coated Fabric, and Sealing Compound
MIL-W-21157	Weldments, Steel, Carbon and Low Alloy
MIL-I-45208	Inspection System Requirements
MIL-M-81807	Marking of Aircraft Ground Support Equipment With Retro-reflective Materials, Process For

STANDARDS

Military

MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-129	Marking For Shipments
MIL-STD-143	Standards, and Specifications, Order of Precedence for
MIL-STD-281	Automobiles, Trucks, Truck-tractors, Trailers and Trailer Dollies: Preservation and Packaging of
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-470	Maintainability Program Requirements (for Systems and Equipments).

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STANDARDS (Cont'd)

Military

MIL-STD-471	Maintainability Demonstration
MIL-STD-756	Reliability Analysis
MIL-STD-781	Reliability Tests: Exponential Distribution
MIL-STD-785	Reliability Program for Systems and Equipment Development and Production
MIL-STD-808	Treating and Painting Procedures
MIL-STD-810	Environmental Test Methods
MIL-STD-889	Dissimilar Metals
MIL-STD-1304	Reliability Report
MIL-STD-1400	Engine Testing Methods
MIL-STD-1472	Human Engineering Design Criteria for Military Systems Equipment and Facilities
MIL-STD-1629	Procedures for Performing a Failure Mode and Effect Analysis for Shipboard Equipment
MS 35000	Battery, Storage, Lead-Acid
MS 90495	Plate, Identification, Blank Mechanical Equipment

HANDBOOKS

MIL-HDBK-472	Maintainability Predictions

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(Copies of specifications, standards and handbooks required by suppliers in connection with specific procurement functions should be obtained from the Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120 or as directed by the contracting officer.)

DRAWINGS

Naval Air System Command

62A94-E3 Base, Enclosure - Turbine, MD-3 Tractor

(Copies of drawings may be obtained from the Naval Air Technical Services Facility, 700 Robbins Avenue, Philadelphia, Pennsylvania 19111 as directed by the contracting officer.)

2.2 Other publications - The following documents form a part of this specification. Unless otherwise indicated the issue in effect on the date of invitation for bids or request for proposal shall apply.

Motor Carrier Safety Regulations

U.S. Department of Transportation, National Traffic Safety Administration Motor Vehicle Safety Standards

(The above regulations are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

Society of Automotive Engineers Handbook

(The above handbook is available from Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA. 15096.)

3. REQUIREMENTS

3.1 First article - The tractors furnished under this specification shall be a product which has been inspected and passed the first article inspections specified herein (see 4.3 and 6.3.1).

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3.2 Selection of specifications and standards - Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials - Materials shall conform to the applicable specifications and shall be as specified herein. Materials not covered by applicable specifications or standards shall be of the best quality and entirely suitable for the purpose intended.

3.3.1 Standard parts - AN, MS, or MIL standard type parts shall be used whenever they are suitable for the purpose, and shall be identified on the drawings by their part numbers. Commercial parts such as bolts, nuts, screws, and washers may be used provided they possess suitable properties and are replaceable by the AN, MS or MIL standard parts without alteration, and provided the corresponding standard part numbers are referenced in the parts list and appear on the contractor's drawing. In the event there is no suitable corresponding standard part in effect on the date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.3.2 Dissimilar metals - The use of dissimilar metals used in contact with each other shall be in accordance with MIL-STD-889.

3.3.3 Rubber - All rubber and similar materials used in the tractor shall be compounded with a reaction product to improve their oil resistance, electrical insulating properties, and ability to withstand extreme temperatures.

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3.3.4 Interchangeability - All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable with each other with respect to installation and performance.

3.4 Design - The tractor shall be designed to have a short compact wheel base suitable for maneuvering in restricted areas and to meet all performance requirements as specified herein. Its function is to tow aircraft and ground support equipment. The design shall be in accordance with MIL-S-8512.

3.4.1 Human Engineering - The requirements of MIL-STD-1472 shall be adhered to in the design of the tractor when considering the operator and passenger compartments and the operational and maintenance requirements to the extent that they are compatible with the requirements of this specification.

3.4.2 Characteristics - The tractor design shall include but not be limited to the following:

(a) Two-wheeled drive and two-wheel steer

(b) Minimum drawbar pull - 7500 pounds

(c) Height - 38 inches maximum

(d) Length - 126±2 inches (less pintles)

(e) Width - 66 inches maximum

(f) Turning radius - 140±2 inches

(g) Ground clearance - 5-1/2 inches minimum

(h) Weight - (less fuel) - 9000 pounds maximum

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3.4.3 Air pollution control - The tractor shall comply with the Environmental Protection Agency Regulations governing the control of Pollution from new motor vehicles and New Motor Vehicle Engines.

3.4.4 Environmental requirements - The tractor shall be subjected to and pass the environmental tests specified in 4.6.13. The tractor shall show no evidence of damage or deterioration after each environmental test.

3.5 Reliability

3.5.1 Reliability Requirements - The tractor shall have a specified Mean Time Between Failure (MTBF) of 250 hours as defined in MIL-STD-781 when tested and accepted in accordance with the provisions of Section 4 and 6 of this specification.

Failure shall be defined as an inability to deliver the specified performance when operating within the environmental conditions of this specification, and any malfunction which the operator or maintenance personnel cannot remedy by adjustment, repair or replacement within thirty minutes, and which causes or can cause (1) cessation of operations or degradation of performance ability of the tractor or subsystem below designated levels; (2) serious damage to the tractor or subsystems by continued operations; or (3) serious personnel hazard.

3.5.1.1 Reliability Program - The contractor shall conduct an approved reliability program in accordance with MIL-STD-785. The reliability program shall be responsive to the general mission requirements together with the specific reliability requirements of this specification.

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3.5.1.2 Program Plan - When specified, a Reliability Program Plan, in accordance with MIL-STD-785 shall be submitted to the procuring activity. The format and content of the Reliability Program Plan shall be in accordance with MIL-STD-785. The Reliability Program Plan, as approved by the procuring activity, will be incorporated into the contract and become the basis for determining contract compliance. The contractor shall include in the Reliability Program Plan methods to insure that results of reliability analysis will produce confirmation of, or changes to, the design. The contractor shall include in the Reliability Program a schedule of the reliability tasks indicating significant program milestones. Provisions shall be included which will assure timely completion and application of interim and final results of each task. (See 6.2.3.)

3.5.1.3 Program Reviews - Scheduled Formal Program Reviews per MIL-STD-785 shall be conducted commencing 30 days after contract award and every 60 days thereafter, and whenever the situation indicates the necessity for additional reviews. Procuring activity personnel shall be notified of such additional reviews at least ten working days prior to the review. An agenda shall be proposed to the procuring activity at the time of notification. The Program Review shall include, but not be limited to, the status of each reliability task, an assessment of the reliability progress, results of the reliability design reviews, results of testing, and resultant planned and implemented design changes.

3.5.1.4 Test Plan - When specified, a reliability test plan, prepared in accordance with MIL-STD-785, shall be submitted (see 6.2.3).

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3.5.1.5 Status Reports - The contractor shall prepare reliability program status reports as described in the CDRL (Contractor Data Requirements List). (DD Form 1423.)

3.5.1.6 Design Reviews - Formal Design Reviews in accordance with MIL-STD-785 shall be conducted in conjunction with the regularly scheduled Program Reviews and whenever the situation indicates the necessity for additional reviews. The design review shall be conducted by contractor personnel independent of the equipment design team. Procuring activity personnel shall be invited to participate. Procuring activity personnel shall be notified of any planned or additional reviews at least ten working days prior to the review. A detailed report of all Formal Design Reviews shall be prepared and submitted in accordance with MIL-STD-785.

3.5.1.7 Subcontractor and Supplier Reliability Programs - The contractor shall be responsible for all subcontractor and vendor reliability programs. Applicable Reliability requirements shall be established for inclusion within subcontractor and supplier procurement documentation.

3.5.1.8 Program Interface - The contractor shall be responsible for interfacing efforts in accordance with MIL-STD-785. The interchange of information between interfacing activities shall be facilitated by use of the format specified in CDRL. (DD Form 1423.)

3.5.1.9 Reliability Analysis - The contractor shall perform a reliability analysis commencing with contract initiation and shall include

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the requirements of 5.2.2 of MIL-STD-785 and subparagraphs thereto. The analysis shall include a reliability apportionment, a mathematical model, and a prediction. The reliability prediction shall be in accordance with MIL-STD-756.

3.5.1.10 Design Analysis - An analysis of the equipment design shall be made by the contractor engineers, independent of the design team. The analysis shall be accomplished on all subassemblies and assemblies of the tractor, electrical, and mechanical. The release of any subassembly or assembly design for fabrication, shall be permitted only with the written concurrence of this independent design analysis group.

3.5.1.11 Failure Modes, Effects, and Criticality Analysis (FMECA) - An FMECA will be conducted in accordance with MIL-STD-1629 and will serve as a tool in further developing the mathematical models, in reapportioning and updating predicted reliability, and determining critical and weak areas. This analysis shall include treatment of all major components of each subsystem. The technique to be used in performing the FMECA shall be described in the Reliability Program Plan. Potential failures which would be mission critical shall be investigated further to determine means of reducing risks to acceptable levels. The contractor shall establish a method for identification, control, and handling of those items that remain critical potential failures following the further investigation and design improvements. Results of the analysis shall be submitted to the procuring activity in an FMECA report to be included in a Status Report.

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3.5.1.12 Failure Reporting, Analysis, and Corrective Action - Procedures for Failure Reporting, Analysis, and Corrective Action shall be included in the Reliability Program Plan as called for by 5.4.1 of MIL-STD-785. The failure reporting system shall ensure that all failures which occur from the start of subassembly and assembly fabrication and test, are recorded and reported to program management. A chart shall be used for presenting this information to the procuring activity. The chart shall be included in the Reliability Status Reports and be continuously updated.

3.5.1.13 Failure Reports - Any and all failures (both relevant and irrelevant) which occur during equipment tests shall be recorded and reported in accordance with the applicable reliability test specification and the Failure Reporting, Analysis, and Corrective Action requirements of the approved Reliability Program Plan. The reports shall include the results of the failure analysis and be included in the status report.

3.5.1.14 Failure Analysis - An analysis of the cause of each failure shall be made in accordance with the applicable reliability test specification and the Failure Reporting, Analysis, and Corrective Action requirements of the approved Reliability Program Plan.

3.5.1.14.1 Mechanical Disparities. Any mechanical disparities observed during test or at the completion of the test require corrective action by the contractor.

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3.5.1.15 Test result report - When specified, a reliability test result report shall be submitted. The content of the report shall be in accordance with MIL-STD-1304 (see 6.2.3).

3.6 Maintainability - The tractor shall be designed and constructed to provide a minimum amount of time necessary for assembly, disassembly, location of trouble sources, and maintenance including servicing. Where practical, parts and components shall be located or positioned for rapid and simple inspection and recognition of excessive wear or potential failure. The configuration shall permit adjustments, servicing, replacement of parts and components, and other maintenance with minimum disturbance to adjacent equipment, parts, or components. Where practical, components that interfere with each other shall be installed to permit greater accessibility to the component predicted to require the most maintenance. The configuration shall permit maintenance with general-purpose tools and equipment normally available commercially. The use of special-purpose tools shall be approved by the procuring activity.

The tractor shall have a specified Mean Time Between Maintenance Action (MTBMA) for scheduled maintenance actions at both the intermediate and organizational level of 150 hours.

The tractor shall have a Mean Time to Repair (MTTR) at the intermediate level of two hours or less. Maximum repair time at the intermediate level shall be four hours.

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Time to repair shall be defined as the time required to return the tractor to operating status, exclusive of transportation time involved.

Daily and pre-operational inspection and adjustments shall be accomplished within 0.25 hours.

Scheduled maintenance actions shall be limited to one man hour per 25 operating hours.

3.6.1 Each maintenance operation listed below shall be accomplished by one man (except as noted) within the time indicated. The operations shall be performed without the use of special tools.

(a) Remove, replace, and adjust fan belts - 30 minutes.

(b) Remove and replace alternator and adjust belt tension - 45 minutes.

(c) Remove and replace regulator - 15 minutes.

(d) Remove and replace fuel filter elements - 10 minutes.

(e) Remove and replace battery - 15 minutes.

(f) Drain engine crankcase, remove and replace oil filter, and refill engine crankcase - 30 minutes.

(g) Drain transmission oil, remove and replace all filter elements and strainers, and refill - 60 minutes.

(h) Remove and replace starter - 30 minutes.

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(i) Bleed and adjust brakes and refill master cylinder (two men) - 45 minutes.

(j) Remove rear lights and pintle from bumper plate and install on the gas turbine enclosure (two men) - 2 hours.

(k) Remove and replace engine coolant system hoses - 30 minutes.

3.6.2 Maintainability Program - The contractor shall develop and implement a Maintainability Program in accordance with MIL-STD-470. The maintainability program shall be responsive to the general mission requirement together with the specific maintainability requirements set forth herein.

3.6.3 Program Plan - When specified, a Maintainability Program Plan shall be submitted to the procuring activity. This Program Plan shall be in accordance with MIL-STD-470 and have paragraph headings similar to Section 5 of MIL-STD-470. The Maintainability Program Plan, as approved by the procuring activity, will be incorporated into the contract and become the basis for determining contract compliance. The contractor shall include in the Maintainability Program Plan methods to insure that results of the analysis will produce confirmation of, or changes to, the design. (See 6.2.3.)

3.6.4 Program Reviews - Scheduled Formal Program Reviews shall be conducted commencing 30 days after contract award and every

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60 days thereafter, and whenever the situation indicates the necessity for additional reviews. Procuring activity personnel shall be notified of such additional reviews at least ten working days prior to the review. An agenda shall be proposed to the procuring activity at the time of notification.

3.6.5 Status reports - When specified, maintainability status reports prepared in accordance with MIL-STD-470 shall be submitted (see 6.2.3).

3.6.6 Subcontractor and Vendor Maintainability Programs - The contractor shall be responsible for all subcontractor and vendor maintainability programs. Maintainability requirements shall be included in subcontractor and vendor procurement documentation.

3.6.7 Commercial Equipments - The contractor shall be responsible for the maintainability of all assemblies and/or equipment purchased as off-the-shelf and/or constructed according to best commercial practice. The contractor shall gather appropriate maintainability assurance data from the commercial vendor, or in lieu thereof, the contractor shall derive the data. In no case, shall the contractor construe his overall system maintainability requirement as excluding the maintainability of commercial equipments.

3.6.8 Program Interface - The contractor shall be responsible for interfacing efforts in accordance with 5.1 of MIL-STD-470. The interchange of information between interfacing activities shall be facilitated by use of the format specified in CDRL (DD Form 1423).

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3.6.9 Maintainability Predictions - Maintainability Predictions shall be in accordance with Procedure IV of MIL-HDBK-472. The contractor shall submit the technique of Maintainability Prediction to the procuring activity for approval along with the Maintainability Program Plan. The prediction technique shall identify the assumed maintainability distribution. A maintainability prediction shall be accomplished within 90 days after award of contract. Additional predictions shall be made whenever a significant design change occurs that would appreciably increase repair time for any subassembly. These predictions shall represent the current state of the design at the time when they are made.

3.6.10 Design Reviews - Formal Design Reviews per 5.9 of MIL-STD-470 shall be conducted by contractor personnel independent of the equipment design team. Procuring activity personnel shall be notified of any planned or additional reviews at least ten working days prior to the review. Detailed reports of all Formal Design Reviews shall be prepared in accordance with 5.9 of MIL-STD-470.

3.7 Logistics

3.7.1 Maintenance - Maintenance of the tractor will be in accordance with OPNAVINST 4790.2A. This instruction provides the basis for the establishment of standard organizations, procedures, and responsibilities for the accomplishment of all maintenance on Naval aircraft, associated material, and equipment. Maintenance of the tractor will be performed by Organizational, Intermediate and Depot Activities.

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3.7.1.1 Organizational Maintenance - The organizational maintenance will be in accordance with OPNAVINST 4790.2A, Figure II-2-16 and NAVAIR 19-600-46-6-1.

3.7.1.2 Intermediate Maintenance - The Aircraft Intermediate Maintenance Departments (NAVY-AIMD) and Headquarters and Maintenance Squadrons (Marine H&MS) are the controlling custodians of aircraft tow tractors. They are responsible to see that all scheduled/unscheduled maintenance is performed in accordance with OPNAVINST 4790.2A, Figure II-2-16 and NAVAIR 19-600-46-6-2. This maintenance level is capable of equipment test, repair adjustment or replacement of damaged or unserviceable parts, components or assemblies.

3.7.1.3 Depot Maintenance - Depot maintenance and repair activities are capable of performing the tasks on material requiring major overhaul, complete rebuilding of components, sub-assemblies, assemblies and end items, including the manufacture or modification of components or assemblies, and testing as required. Depot maintenance serves to support lower echelons of maintenance by performing that maintenance beyond their capability.

3.8 Engine - The engine shall be commercially available, liquid cooled, eight cylinder configuration, and shall have a piston displacement of not less than 260 cubic inches and a torque of not less than 200 foot-pounds at governed rpm. Engine manufacturer's published ratings for engine shall not be arbitrarily raised to meet the require-

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ments specified herein. Engine torque and horsepower at governed rpm shall develop 7500 lbs of pull with a 20% safety margin. Engine shall be designed to operate on low leaded gasoline per VV-G-001690. An engine overheat relay shall be provided to indicate engine overheat by actuating a red light visible to the tractor operator.

3.8.1 Engine specifications - Engine specifications including engine torque, speed, displacement, horsepower curve and related data shall be submitted by the supplier (contractor) as part of the bid or proposal. The specifications shall be the engine manufacturer's specifications.

3.8.2 Safety device - The engine shall contain a safety device that will automatically stop the engine when the engine coolant temperature exceeds the engine manufacturer's high temperature coolant limit.

3.8.3 Governor - The engine shall be equipped with a governor which shall be set to insure that the maximum tractor speed is 15 mph. It shall be sealed in a manner that will reveal tampering with the governor setting.

3.8.4 Air cleaner - An air cleaner shall be furnished and mounted on the right front side of the engine compartment. The unit shall be of the oil bath type built to filter air into the carburetor by passing it over or through an oil bath and finally through oil wetted mesh to trap dust particles and contaminants.

3.8.5 Throttle linkage - The throttle linkage shall consist of a mechanical linkage with a spring to obtain positive throttle con-

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trol and preclude erratic acceleration of the tractor. A cable linkage is not acceptable. The control linkage shall conform to Motor Vehicle Safety Standard (MVSS) 124.

3.8.6 Accelerator pedal - The accelerator pedal shall be located within easy reach of the seated operator. It shall be so designed that when the heel of the operator's heavy boot is resting on the floor, his toe will not encounter any interference from the firewall.

3.8.7 Transmission - The transmission shall be a heavy duty automatic planetary full torque shifting unit equipped with a hydraulic torque converter. The transmission shall have not less than three forward speeds, one reverse speed, and one neutral position. When placed in high gear it shall automatically go through the lower gears. Input torque capacity shall be not less than the engine and torque converter maximum output torque. An external oil cooler shall be provided to maintain proper oil temperature under severe operating conditions and an overheating warning device shall be incorporated. The transmission shift lever mechanism shall conform to MVSS 102 except that a park position on the controls is not acceptable. A stop shall be installed on the transmission shift mechanism to prevent unintentional shift lever movement from forward drive into reverse unless the reverse release button is depressed. The transmission bands shall be easily accessible and readily adjustable.

3.8.8 Axle - The drive axle shall be of double reduction gear type. The final reduction shall be in the outer wheel hub.

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3.9 Electrical system - The tractor shall be provided with a complete electric starting and lighting system, amply equipped with safety and regulating devices. Adequate fuses shall be installed for the tractor ignition system, headlights, tail lights, and the stop lights. All control switches shall be within convenient reach of the driver. Ground polarity of the system shall be negative. Lighting requirements shall conform to U.S. Department of Transportation regulations, paragraphs 393.24(a) and (d), 393.25(d) and 393.25(f). Lamps and associated hardware shall conform to MVSS 108 unless otherwise specified. The design of the electrical system shall include quick disconnect connectors at the instrument panel. Wiring end connections shall be coated with a sealing compound conforming to class 1, 2, or 3 of MIL-S-8516.

3.9.1 Ignition system - The tractor shall be equipped with a 12-volt ignition system suitably protected by fuses. The system shall include switch, ammeter, ignition coil, high tension cables, low tension cables, and spark plugs. Grommets shall be provided whenever wiring passes through partitions and structural members.

3.9.1.1 Ignition coil - The ignition coil shall be mounted in a position isolated from engine heat and the exhaust manifolds.

3.9.1.2 Ignition switch - A keyless ignition switch shall be furnished. It shall have not less than two positions, "on" and "off", legibly marked.

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3.9.1.3 Distributor - The distributor shall be equipped with a fully automatic centrifugal governor to advance and retard the spark in accordance with automotive practice.

3.9.2 Starting system - The starting system shall consist of a starting motor, solenoid relay switch, neutral safety switch, starter button, all necessary wiring, and heavy duty cables properly assembled in a complete unit.

3.9.2.1 Starter - The starter motor shall be 12-volt, of adequate capacity, substantially mounted, and equipped with a pinion-meshing device or over-running clutch. The starter motor shall be effectively protected from exhaust heat. The starter shall be of the soft start type or Bendix drive type and shall not turn until after engagement with the engine ring gear.

3.9.2.2 Starting motor switch - The starting motor switch shall be a solenoid type, connecting the starting motor to the battery. It shall be of a design to preclude starter engagement with the ring gear while the engine is operating.

3.9.2.3 Neutral safety switch - A neutral safety switch wired in series with the starting circuit shall be connected to the shift lever by a suitable arrangement that will allow the starting circuit to open and remain open when the shift lever is moved out of the neutral position.

3.9.3 Alternator - The alternator shall be 12-volt, 80 ampere capacity and shall start charging the battery at engine manu-

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facturer's recommended engine idling speed. The alternator shall be serviced from the top side of the tractor and shall be located for easy removal without disturbing any other component. The alternator shall be protected or isolated from engine heat.

3.9.4 Battery - The tractor shall be equipped with a 12-volt battery conforming to MS 35000 and MIL-B-11188, Type 6TN.

3.9.4.1 Battery mountings - The battery shall be in an easily accessible compartment to facilitate removal. If mounted outside the engine cover the battery shall be housed in an individual box fitted with a quick release cover to provide easy inspection and servicing. Covering and positioning shall adequately protect against short circuit contact of electrical wiring and battery straps.

3.9.4.2 Acid proofing - All surfaces within 12 inches of the battery and all parts subject to acid spillage shall be treated with two coats of acid resistant lacquer conforming to TT-L-54.

3.9.5 Lighting system - The tractor lighting system shall include headlights, combination tail and stoplight, rear pintle light, and an instrument panel light. All lights shall operate from the 12-volt battery supply. Adequate fuses shall be installed for the headlights and the combination tail and stoplight.

3.9.5.1 Headlights - The tractor shall be equipped with two headlights of not less than 32-candle power each. Lamps shall be equipped with wide dispersion type lens, shall be adequately recessed behind

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the front bumper to prevent damage, and shall be vertically and horizontally adjustable in accordance with automotive practice. The "ON-OFF" control switch shall be mounted on the instrument panel.

3.9.5.2 Combination tail and stoplight - One red reflector type combination tail and stoplight of not less than 3-candle power and 15-candle power respectively shall be mounted on the rear of the tractor. It shall be adequately recessed to protect it against damage. The stoplight shall operate automatically upon actuation of the service brake, including when the ignition switch is in the "OFF" position.

3.9.5.3 Rear pintle light - One clear light of not less than 15-candle power shall be mounted on the rear of the tractor for illuminating the rear pintle. This light shall be adequately recessed to protect it against damage. The controlling "ON-OFF" control switch shall be mounted on the instrument panel.

3.9.5.4 Dash panel light(s) - The illumination of instruments and controls on the dash panel shall be controlled by a rheostat with an "OFF" position. The dash panel light(s) shall provide for good readability of all gages, meters, controls, and instructions on the panel.

3.9.5.5 Instrument panel - The instrument panel shall contain but not be limited to the following:

Engine Temperature Gage

Engine Temperature Warning Light (Flashing)

Torque Converter Temperature Gage

Torque Converter Temperature Warning Light

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Oil Pressure Gage

Oil Pressure Warning Light

Light Switches

Starter Switch

Ignition Switch

Dash Panel Light

Hour Meter

Ammeter

Fuel Gages (tractor and turbine fuel tanks)

Parking Brake "ON" Warning Light (Flashing)

Horn

All instruments and switches shall be waterproof, moistureproof, and dustproof. Each instrument and switch shall be identified by an indented or embossed metal plate fastened to the panel by two metal screws, or waterproof adhesive. The material for identification plates shall conform to MIL-P-514. All instruments shall be mounted flush with the panel. A clear plexiglas shield shall be mounted over the glass encased instrument gages.

3.9.5.6 Engine temperature gage - The engine temperature gage shall be accurate to within 5°F at 180°F.

3.9.5.7 Engine hour meter - The engine hour meter shall have a time totalizing mechanism of not less than 9,999 hours. It shall be connected in series with the ignition switch and engine lubrication pressure switch. The meter shall be constructed to insure continuous trouble free performance under severe operating conditions.

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3.9.5.8 Horn - The tractor shall be equipped with an electric horn. The moistureproof horn button shall be Cole-Hersee Co., 20 Old Colony Avenue, Boston, Mass. 02127; Part No. 9245 or equal.

3.9.5.8.1 Location - The horn actuating button shall be located on the instrument panel. It shall be positioned on the driver's side of the panel approximately 2 inches from the top and 2-1/2 inches from the edge of the panel. An actuating button or tilt ring located on the steering wheel is not acceptable.

3.9.5.9 Protective cover - A protective cover shall be placed over the actuating button. The cover shall be in accordance with Naval Air Systems Command Drawing No. 65A81C281-2 or equivalent.

3.10 Fuel system - The fuel system shall consist of a fuel tank with sediment bowl, fuel pump, carburetor, connecting fuel lines and a turbine fuel tank.

3.10.1 Tractor fuel tank - The tractor shall have a fuel tank of not less than 20 gallon capacity. The tank shall be equipped with a filler tube, a float type rheostat, baffles (to prevent fuel sloshing), a shut-off cock with fuel line connection, a vent, an easily accessible drain plug, and an overflow line to carry excess fuel away from the engine, muffler, and tail pipe. The filler opening, sufficient to receive a 1-1/2 inch diameter filling nozzle, shall be easily accessible. A chain secured filler cap shall be provided. The cap shall be painted red with "GAS" painted in white letters across the top per paragraph 3.23. The low lead gasoline used in this tank shall conform to Specification VV-G-001690.

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3.10.2 Fuel pump - The fuel pump shall be a diaphragm type actuated by a rocker fork arm and an eccentric on the cam shaft or a commercial electric fuel pump. Moving parts of the diaphragm type pump shall be lubricated through the opening in the cylinder block.

3.10.3 Carburetor - The carburetor shall be of the aspirating or injection type, properly balanced to maintain correct air and fuel mixture in accordance with automotive practice. Provision shall be made for unobstructed adjustment of the idle metering system and the idle limit stop. The main metering system shall be controlled by removable fixed orifice jets designed for sea-level operation.

3.10.4 Fuel lines - Adequate fuel lines shall be provided to insure optimum operation at all speeds. The lines shall be protected from engine heat by suitable insulation or heat reflectors.

3.10.5 Turbine fuel tank - The turbine fuel tank containing baffles shall be constructed of aluminum alloy or stainless steel. It shall have a capacity of 50 gallons. The filler opening shall be easily accessible. It shall be designed to receive a 2-1/2 inch diameter filling nozzle, and shall be provided with a chain secured cap. The cap shall be painted white with "JP-5" painted in red letters across the top per paragraph 3.19. When in military service, turbine fuel per Specification MIL-T-5624, Grade JP-5 will be contained in this tank.

3.10.5.1 Fuel connection - The fuel line for supplying fuel from the turbine fuel tank to the gas turbine shall have a shut-off cock with a male connector (E.B. Wiggins Inc., 5000 Triggs St., Los Angeles,

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CA. 90022, Part No. 6005E8 or equal) at the rear of the tractor for connecting to the gas turbine (GTC-85). The connector shall not protrude beyond the rear of the tractor when the Gas Turbine Enclosure is not installed.

3.11 Cooling system - The cooling system shall consist of a radiator with a filler cap, water pump, fan and fan belt, thermostat and connecting hose and fittings. Two drain cocks shall be provided. One drain cock shall be located in the base of the radiator (or pump inlet elbow if at the low point) and the second outlet shall be located at the lowest point of the liquid coolant jacket in the cylinder block. The system shall contain sufficient volume to prevent coolant temperature from exceeding 200°F in an open system or 225°F in a pressurized system as measured at the head outlet when operating continuously under normal conditions in an ambient temperature of 125°F. Pressurized systems shall be provided with a positive operating pressure relief valve opening at 7 psi or less, and a positive operating vacuum breaker valve opening at a pressure differential of one (1) psi or less.

3.11.1 Radiator - The radiator assembly shall consist of a fin-and-tube or honeycomb type core with expansion tank at the top and sediment tank at the bottom. It shall be protected by a front bumper plate. A surge tank shall be provided.

3.11.2 Water pump - The water pump shall be the centrifugal type equipped with a corrosion resistant steel shaft. It shall contain a lifetime lubrication system or an easily accessible lubrication fitting.

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3.11.3 Thermostat - A nominal 180°F thermostat shall be located in the engine head outlet to the radiator and shall be easily removable.

3.12 Exhaust system - The exhaust system shall consist of an exhaust manifold and pipe connected to a muffler and a tail pipe. The exhaust system shall be designed and constructed to protect personnel from burns.

3.12.1 Muffler - The muffler shall be a spark arresting muffler. It shall be capable of withstanding 20 engine backfires without damage to the muffler.

3.12.2 Tail pipe - The tail pipe shall not extend beyond the end of the tractor and shall be secured with adequate brackets. The exhaust gases shall be emitted in a horizontal direction at the rear of the tractor.

3.13 Steering - Steering shall be accomplished by an automotive type steering wheel mounted on a steering column. The steering wheel shall turn the front wheels by worm and sector or cam and lever, with power assist reduction device which shall permit easy operation. Clockwise or right hand rotation of the steering wheel shall provide for right hand or clockwise turning of the tractor while in a forward motion. Provision shall be made for adjustment to compensate for wear of all major steering parts, particularly steering gear reduction mechanism. All friction points shall contain an easily accessible lubrication fitting or lubricant reservoir. When provided, the reservoir shall be

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readily accessible for checking the lubricant level. The steering wheel shall be capable of being turned from extreme right to extreme left or vice versa by the operator exerting a force not greater than 10 pounds at the rim while the unloaded tractor is stationary on a dry concrete surface.

3.14 Suspension system - The front axle shall be suspended on leaf springs. The suspension system shall support static loads without damage or deformation to the suspension system or to the engine transmission or tractor chassis. The wheel mounting shall be three or four point suspension and shall include wide tread knuckle-type steer. The suspension system shall be designed to prevent a vertical movement of the rear pintle in excess of two inches when starting or stopping the tractor under load or no load conditions.

3.15 Wheels - The tractor shall be mounted on six (6) steel, malleable iron, or cast iron wheels, equipped with tapered roller bearings. Front wheels, when equipped with new tires, shall clear any part of the tractor structure by not less than 1/2 inch. Tire chain clearance for dual rear drive wheels shall be not less than clearances specified in SAE J683.

3.16 Tires - The pneumatic tractor tires shall be as specified in MIL-T-12459, Class MS, Style (a) 8 ply. The front tires shall be 6.00 x 9. The rear tires shall be 7.50 x 16.

3.16.1 Tubes - The tires shall be equipped with tubes conforming to the requirements of MIL-T-12459.

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3.17 Brakes -

3.17.1 Service brake system - The tractor shall be equipped with a split-service, power brake system of the hydraulic, internal expanding type, capable of safely stopping, holding and controlling the movement of the tractor under the operating conditions and in the environments specified herein. A brake pedal pressure of approximately 26 pounds shall be sufficient to apply brakes to bring the tractor to a complete stop in not more than 30 feet from a speed of 15 MPH. When excessive foot pressure is applied to the brake pedal, all force shall be transmitted to the master cylinder without causing any deflection or bending to the brake linkage or mounting structure. All other brake system components shall be protected from damage or malfunction and shall provide maximum safety and ease of maintenance. The service brake system shall include all the components required for a complete, matched brake system. The wheel brake assembly rated torque capacity, heat rejection characteristics, and installation shall be in accordance with the recommendations of the wheel brake assembly manufacturer. The fixed assemblies shall be positively indexed to the spindle assembly. A dual master cylinder shall be furnished with one master cylinder supplying fluid to the rear wheel brake on the driver's side. The other master cylinder shall supply fluid to the rear wheel brake on the passenger side. The hydraulic master cylinders shall be accessible for checking and refilling without removal of major components. A power booster is ac-

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ceptable. When a vacuum booster is furnished, the reservoir shall have sufficient capacity for a minimum of three full service brake applications with the engine off. Brake lining for shoes or pads shall be for heavy-duty service and shall be securely bonded or riveted to the brake shoe. Manual brake adjustment shall be provided. The brakes shall be adjustable using common handtools or a star wheel brake tool. Wheel cylinders shall be located so that hydraulic lines may be bled of air without the necessity of removing the wheels or disconnecting hydraulic lines. The service brakes, except for stopping distance requirements and as otherwise specified herein, shall conform to MVSS 105-75.

3.17.1.1 Tubing and hoses - The tubing used in the service brake system shall be corrosion resistant steel conforming to the requirements of MIL-T-8504. The hoses in the service brake system shall conform to MVSS 106-74.

3.17.2 Parking brake system - The parking brake shall be an independent mechanical friction brake. It shall be hand operated and equipped with a locking device. If the hand lever is of the ratchet type, the ratchet and pawl shall be of heat treated steel. The brake system shall be mounted on the drive or propeller shaft rotating with the wheels and consist of either an internal-expanding or an external-contracting type mechanical friction brake. It shall be capable of holding the tractor on a 36 percent grade. The hand lever shall be located within reach of the operator in a noninterfering position, permitting safe movement on and off the tractor. Adjustment of parking brake shall be accomplished without the removal of any major assembly other than tires and wheels.

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3.18 Main frame - The frame and related structure shall be fabricated from structural steel or steel castings and shall be sufficiently rigid to prevent any distortion. Frame joints shall be adequately riveted or welded to provide a unit structure.

3.19 Body - Tractor body shall be constructed of heavy steel sheet or plate equal to that used in the manufacturer's normal commercial product and shall enclose the engine and all parts requiring protection. A waterproof, inner spring, type, cushion seat with back rest and an outer arm shall be located directly behind the steering wheel. This seat shall be adjustable fore and aft. The right side of the tractor shall be equipped with a similar seat. The side panels and top deck shall be constructed of steel not less than 0.250 inch thick. A hand grab shall be mounted within easy reach of the passenger crew member on the right side of the tractor. The hand grab shall be suitable for grabbing by a man wearing heavy mittens. The fastener centers shall be approximately 10 inches apart.

3.19.1 Fenders and floor boards - The tractor shall be equipped with fenders over all wheels. The rear fenders shall be steel plate or cast iron. The floor shall be constructed of steel not less than 1/4 inch thick and shall be covered with Black Safety Walk, conforming to MIL-D-17951, medium, type B.

3.19.2 Bumper plates - Bumper plates in accordance with MIL-M-8090 shall be provided at both ends of the tractor. The rear bumper plate shall have thirteen (13) 5/8-18UNF tapped mounting holes as shown on Naval Air Systems Command Drawing Number 62A94-E3. The rear bumper plate shall also house the rear lights and pintle.

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3.19.2.1 The rear lights and pintle shall be capable of being removed from the bumper and installed on the gas turbine fuel tank enclosure, when used.

3.19.3 Cowl and hood - The engine shall be completely enclosed by a rounded cowl and a one-piece hood hinged at the front end. The hood shall be equipped with mechanical restraints to allow it to be secured in the raised position for maintenance and servicing. Also, the hood shall be equipped with latches to prevent accidental opening of the cowl when the tractor is in motion.

The cowl and hood shall be constructed of steel plate not less than 0.135 inch thick reinforced with bracing to support a load of 300 lbs./sq.ft. without permanent deformation. The sides of the cowl or hood shall contain adequate louvers for engine cooling. The hood shall incorporate a spring assist mechanism so that an average man can raise or lower the hood from either side with one hand.

3.19.4 Counterweights - Provisions shall be made in the rear of the tractor for two (equal weight) left and right, counterweights. The counterweights shall be readily removable to permit installation of the gas turbine. Alternatively, a single detachable counterweight may be used.

3.19.5 Pintles - Pintles shall be Holland Hitch Company, Holland, Michigan, 49423, Model 400-FA or equal. The front pintle centerline shall be 16 inches above the ground level and the rear pintle centerline shall be 12 inches above the ground level.

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3.19.6 Hoisting and tiedown - The tractor shall be provided with four lifting eyes (2 forward and 2 aft), properly located to allow lifting tractor in a normal position by means of a cable sling without spacer bars. The eyes shall have an inside diameter of not less than 3 inches, and each eye shall be capable of withstanding a load of not less than 500 percent of the gross tractor weight at the respective point of attachment. The notation "Lift Here" shall be stencilled in close proximity to each lifting eye. The eyes shall serve for both hoisting and tiedown.

3.20 Welds - All welding shall be in conformance with MIL-W-21157, Class 3.

3.21 Lubrication fittings - The lubrication fittings shall be in accordance with MIL-F-3541. The fittings shall be provided in accessible positions at all necessary lubrication points. Pressure relief fittings shall be provided where use of high pressure lubrications equipment could cause damage.

3.22 Safety - Provisions for the safety of personnel and equipment shall be incorporated to the maximum extent possible with regard to anticipated operation, maintenance, and storage environments and to the capability of the operating personnel.

3.22.1 Safety Markings - Precautionary markings shall be provided as necessary to warn personnel of hazardous conditions and precautions to be observed to insure the safety of personnel and equipment.

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A warning note shall be placed adjacent to the front pintle hook in one inch red letters to read "WARNING - USE CAUTION WHEN MANEUVERING ACFT WITH FWD PINTLE".

3.23 Finishes and Protective Coatings - All protected metal parts and surfaces that are constructed from corrosion-resistant material and on which a protective coating would be detrimental to proper functioning, need not be given protective coatings. All exposed metal parts and surfaces and all protective metal parts on which a protective coating will not be detrimental to proper functioning shall be thoroughly cleaned and painted yellow in accordance with MIL-STD-808, F103DG, Type I. The tractor fuel tank spout and cap shall be painted red in accordance with MIL-STD-808, F103BG, Type I. The turbine fuel tank cap shall be painted white in accordance with MIL-STD-808, F103NG, Type I. Hardware such as latches, clamps, pivots, and others subject to abrasive action either by rotation, rolling or scraping shall be suitably treated to prevent corrosion and malfunction.

3.24 Air transportability - The tractor shall conform to the air transportability requirements of MIL-A-8421.

3.25 Torque coverter fluid temperature - The torque converter fluid temperature shall not exceed 250°F when the tractor is subjected to the high temperature test specified in 4.6.13(b).

3.26 Electromagnetic interference requirements - The tractor shall conform to the electromagnetic interference requirements specified in MIL-STD-461, Category III Class D.

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3.27 Reflective tape - Reflective tape shall be applied along the extreme perimeter of the tractor as specified in MIL-M-81807. The tape shall be four inches wide; color shall be white.

3.28 Identification plate - The identification plate shall be as specified in MIL-P-514, Type I, Style 1. The marking shall be as specified on MS90495. The plate shall be permanently affixed to the personnel compartment side of the fire wall in a conspicuous place.

3.29 Instruction plate - The instruction plate shall be as specified in MIL-P-514, Type I, Style 1. In lieu of the marking specified on MS90495 the plate shall contain concise legible instructions relative to operating the tractor in a safe and efficient manner. The plate shall contain the technical manual number. It shall be permanently affixed on or near the instrument panel.

3.30 Servicing and adjusting - Prior to delivery of the tractor the contractor shall service and adjust each vehicle for operational use including the following: focusing of lights; adjustment of the engine, electrical and brake system; filling and charging of battery; alignment of front wheels; inflation of all tires; complete lubrication of chassis, engine, and running gear, with grades of lubricants recommended for the ambient air temperature at the delivery point; servicing of cooling system with a solution of ethylene glycol and water in equal parts by volume.

3.31 Workmanship - Workmanship shall be of a quality consistent with the highest existing tractor production standards and prac-

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tices. All finished surfaces shall be protected against corrosion or damage prior to delivery. All fins and other excess material shall be removed from castings and forgings. All surfaces, including threads, shall be free from burrs and sharp edges. All material shall be sound, of uniform quality and condition, and free from seams, cracks, and other defects which may adversely affect the strength, endurance, or wear resistance of the part. Welding or other means for the repair of defects in materials shall not be performed unless specifically authorized by the Contracting Officer.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier (contractor) is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 The contractor shall provide and maintain an inspection system in accordance with MIL-I-45208.

4.1.2 Inspection terms - MIL-STD-109 shall govern the definition of terms used herein.

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4.2 Classification of inspections - The inspections of the tractor shall be classified as follows:

(a) First article inspection (see 4.3 and 6.3.1)

(b) Quality conformance inspection (see 4.4 and 6.3.2)

4.3 First article inspection - The first article inspection shall consist of the following examinations and tests:

Visual examinations (4.6.1)

Engine test (4.6.2)

Operation test (4.6.3)

Speed test (4.6.4)

Turning Radius test (4.6.5)

Drawbar pull test with counterweights (4.6.6)

Drawbar pull test with turbine installed (4.6.7)

Start test (4.6.7.1)

Endurance test (4.6.8)

Torque converter test (4.6.9)

Maintenance test (4.6.14.2)

Environmental tests (4.6.13)

Muffler test (4.6.10)

4.3.1 First article sample - Unless otherwise specified, as soon as practicable after the award of the contract or order the manufacturer shall submit one tractor for inspection to determine conformance with this specification (see 6.3.1). This unit shall be representative

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of the construction, workmanship, components, and materials to be used during production. When a manufacturer is in continuous production of these units from contract to contract, submission of a first article sample on the new contract may be waived at the discretion of the Government (see 6.2.2.1). Approval of the first article sample, or the waiving of the first article inspection is not a waiver of the requirement of submitting to the quality conformance inspection.

4.3.1.1 Upon completion of the first article inspection, all of the applicable inspection and test reports, and when applicable, recommendations and comments pertinent for use in monitoring production shall be forwarded to the cognizant government activity (see 6.2). The approved unit shall be returned to the manufacturer for use in monitoring production. It shall be delivered as a unit under the contract.

4.4 Quality conformance inspection - The quality conformance inspection shall consist of the following examinations and tests:

Visual examinations (4.6.1)

Functional tests (4.6.11)

4.4.1 Inspection sampling - The quality conformance examinations and tests shall be performed on each tractor offered for acceptance under the contract or order.

4.5 Test conditions -

4.5.1 Temperature and pressure - Unless otherwise specified, tests shall be conducted at an ambient temperature of approximately 25°C ±5° and a barometric pressure of approximately 29.92 in. Hg.

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4.6 Inspection methods -

4.6.1 Visual examinations -

4.6.1.1 Fluid levels - The fluids (coolant, engine oil, hydraulic oil, transmission oil) shall be examined to insure they are at the proper operating levels.

4.6.1.2 Gages - All gages shall be examined to determine conformance to the requirements of GG-G-76. The fuel gages shall be checked for accuracy at full, half full, and empty levels.

4.6.1.3 Lubrication fittings - Lubrication fittings shall be examined to insure that the tractor has been completely lubricated, and for compliance with the following:

(a) That the tractor is designed for lubrication by standard military lubricants.

(b) That the fittings are as specified in 3.21.

(c) All fittings easily accessible.

(d) Fittings in vulnerable positions are protected by recessing or bosses.

4.6.1.4 Servicing and adjusting - Each tractor shall be examined to determine conformance to the servicing and adjusting requirements specified in 3.30.

4.6.1.5 Automatic transmission bands access - The tractor shall be inspected to determine that no obstacles interfere with the adjustment of the automatic transmission bands.

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4.6.1.6 Inspection of preparation for delivery - Each tractor shall be examined to determine compliance with the requirements of Section 5 of this specification.

4.6.1.7 Each tractor shall be examined to determine conformance to requirements not covered by specific examinations or tests.

4.6.2 Engine test - The engine shall be tested in accordance with MIL-STD-1400, test method 2100, 2200, 2300, 2400, 2500.

4.6.3 Operation test -

4.6.3.1 Test course - The operation test shall be conducted on a course similar to Figure I. The figure is descriptive and shown for convenience of identification. It is not intended to exclude any course that is suitable for the test.

The test shall be conducted as follows:

(a) Couple the tractor to a trailer containing 75000 lbs.

(b) Position the tractor at the starting point indicated in Figure I.

(c) Drive the tractor to position A at a speed not less than five mph.

(d) Stop the tractor by applying the service brake.

(e) Hold tractor in this position for ten seconds.

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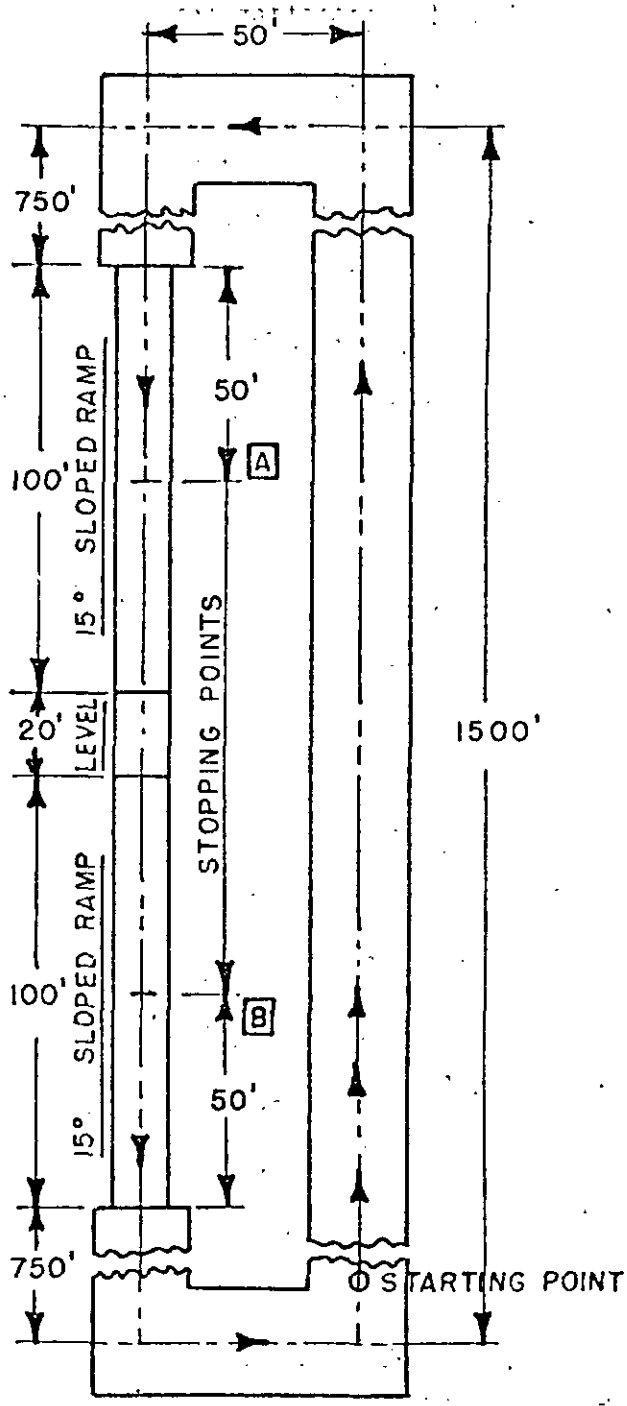


FIGURE I. TEST COURSE

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(f) Apply the parking brake and release the service brake.

(g) After ten seconds release the parking brake and continue to drive the tractor.

(h) Stop the tractor at point B by applying the service brake. Hold the tractor in this position for ten seconds.

(i) Apply the parking brake and release the service brake.

(j) After ten seconds release the parking brake and continue to drive the tractor.

(k) Stop the tractor at the starting position.

Repeat the operatiors test for one hour.

There shall be no sign of instability; damage to the tractor or malfunction of any component part during the test.

4.6.4 Speed test - The fully equipped tractor, without load, shall be capable of maximum speed of 15 mph at an engine RPM that does not exceed the peak horsepower RPM.

4.6.5 Turning radius test - The tractor shall be driven in full 360 degree turns to the right and to the left. The turning radius shall be within 140 ±2 inches measured to the center of the outside wheel tread.

There shall be no sign of instability; damage to the tractor or malfunction of any component part during the test.

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4.6.6 Drawbar pull test with counter weights - On a dry level concrete surface the tractor shall develop a minimum continuous drawbar pull of 7500 pounds at a speed of approximately 1-1/2 mph for 10 seconds. This test shall be conducted not less than four times and the average drawbar pull shall be not less than 7500 pounds.

The tractor shall successfully complete the test and there shall be no signs of instability or damage to the tractor or malfunction of any component part during the test.

4.6.7 Drawbar pull test with turbine installed - With the counterweights removed, the turbine fuel tank empty, and the Government furnished GTC-85 and the enclosure installed, the tractor shall develop a minimum continuous drawbar pull of 7500 pounds at a speed of approximately 1-1/2 mph for 10 seconds.

The tractor shall successfully complete the test and there shall be no signs of instability or damage to the tractor or malfunction of any component part during the test.

4.6.7.1 Start test - The tractor with the turbine installed shall be subjected to four fast starts to determine that the front wheels do not lift off the ground under these operating conditions.

There shall be no failure or malfunction of any component during the test.

4.6.8 Endurance test - The tractor, with a towed load of 52,500 pounds, shall be driven over a smooth paved road at not less than 10 mph for a period of not less than four hours. The use of a heavy

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trailer equipped with dual wheels and pneumatic tires may be used for performing this test. During this operation observe gages for overheating, pressure gages for low pressure, ammeter for low charge rate, and note unusual noises and operating deficiencies. The coolant temperature shall not exceed 200°F in an open system or 225°F in a pressurized system, as measured at the head outlet when operating continuously in an ambient temperature of 125°F. For testing purposes, and to avoid the necessity of operating in a 125°F ambient, the difference between the actual ambient and 125°F shall be appropriately added or subtracted from the coolant temperature when operating with the thermostat removed and using the coolant temperature taken at the radiator expansion tank.

There shall be no malfunction of the tractor and no failure of any component during the test.

4.6.9 Torque converter test - Immediately following the test outlined in paragraph 4.6.8 the vehicle torque converter fluid temperature shall be recorded. With the vehicle transmission in maximum reduction, hold tractor from moving with vehicle brakes, accelerate engine to full throttle. The vehicle shall be held in a stationary position, with engine at full throttle for a period of one (1) minute. The converter oil temperature shall not exceed 250°F in ambient temperature of 125°F. For testing purposes, and to avoid the necessity of operating in a 125°F ambient, the difference between the actual ambient and 125°F will be appropriately added or subtracted.

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There shall be no malfunction of the tractor and no failure of any component during the test.

4.6.10 Muffler test - The muffler shall be subjected to 20 deliberate backfires (see 3.12.1).

4.6.11 Functional tests -

4.6.11.1 Service brake test - The service brake shall be capable of bringing the tractor to a full stop from 10 mph within 20 feet and from 15 mph within 30 feet on dry level concrete. A brake "fade-out" test consisting of bringing the tractor from 15 mph to a full stop 20 times in 30 minutes and then immediately, under maximum acceleration, bringing the tractor to 15 mph and stopping within 80 feet.

4.6.11.2 Transmission test - The transmission test shall be conducted as follows:

(a) Couple the tractor to a trailer containing 30,000 lbs.

(b) Position the tractor at the starting point shown in Figure I.

(c) Shift the transmission through all forward positions while traversing the course.

(d) Stop the tractor at position a; put the transmission in reverse gear, and travel 25 feet in reverse.

(e) Stop the tractor.

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(f) Shift the transmission from reverse to drive position, and attain a speed of 15 mph.

(g) Shift the transmission from the highest to the lowest forward position without decreasing the speed.

There shall be no malfunction or damage to the transmission during the test.

4.6.11.3 Parking brake test - The parking brake test shall be conducted as follows:

(a) Drive the tractor up a grade of 36°.

(b) Apply the service brake.

(c) When the tractor has stopped apply the parking brake and release the service brake.

4.6.12 Electromagnetic radiation test - The tractor shall be tested for radiated electromagnetic interference in accordance with the provisions specified in MIL-STD-461, Category III, Class D. The radiated interference shall not exceed the applicable limits or the tractor shall be considered to have failed the test.

4.6.13 Environmental tests - Using the test methods specified in MIL-STD-810. The tractor shall be subjected to and pass the following environmental tests:

Test conditions:

There shall be no protective covering placed over the tractor during the tests.

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Pretest data, as specified in MIL-STD-810, is required for each test.

(a) Altitude - test method 500 - procedure I.

(b) High temperature - test method 501, procedure II. Internal chamber temperature shall be 125°F. In addition, the temperature of the torque converter fluid shall be measured and recorded (see 3.25).

(c) Low temperature - test method 502, procedure I; step 2 shall be conducted at -65°F for 24 hours; step 4 temperature shall be 0°F.

(d) Sunshine - test method 505, procedure I.

(e) Rain - test method 506, procedure I.

(f) Humidity - test method 507, procedure I.

(g) Fungus - test method 508, procedure I.

(h) Salt fog - test method 509, procedure I, in addition to the requirements of procedure I the tractor shall be inspected and operated after 24 hours of exposure.

(i) Dust - test method 510, procedure I. The tractor shall be inspected and operated immediately after the test.

4.6.14 Reliability and Maintainability Demonstration - The test requirements of this section are for reliability and maintainability

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and do not replace design, performance, environmental, miscellaneous or other tests specified for the tractor.

4.6.14.1 Reliability Demonstration - The contractor shall perform a Reliability Demonstration Test Program in accordance with MIL-STD-781 for one tractor. (Two or three tractors may be used to reduce the testing period.)

4.6.14.1.1 Test Procedure - The contractor shall prepare a detail test procedure to accomplish the Reliability Demonstration Test program for approval by the procuring activity in accordance with MIL-STD-781. The procedure shall include: the facilities required to perform the tests, articles to be tested; a test schedule with expected decision points denoted; the environmental conditions and cycling schedule; measurement endpoints; relevant failure definitions; in addition to the requirements of 5.1.3 of MIL-STD-781. These tests will be performed at the contractor's facilities.

4.6.14.1.2 Accept/Reject criteria - Accept/reject criteria for the Reliability Demonstration Test shall be in accordance with MIL-STD-781, paragraph 4.2.8.8, Test Plan No. 8, Test Level A-1 with on/off cycling. The length of test shall conform to Test Plan No. 8 requirements. The total unit hours of test and unit hours per tractor shall be continuously monitored by the contractor. The Government reserves the right for a Government observer, representing the Contracting Officer, to be present throughout the conduct of the acceptance tests. Unit hours of test from endurance testing can be applied to reliability testing.

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4.6.14.1.3 Reliability Test Reports - Reliability Test Reports shall be in accordance with MIL-STD-781.

4.6.14.2 Maintainability Demonstration Test - A maintainability demonstration test shall be conducted on the tractor to evaluate its design limitations in terms of MTTR. The test shall be in accordance with MIL-STD-471, Method 1, Plan A_1 and B_1 and shall include corrective maintenance as well as preventive maintenance tasks. The tractor shall demonstrate an MTTR of 2.5 hours or less. The maintenance operations listed in paragraph 3.6.1 shall be included in the demonstration.

4.6.14.2.1 Maintainability Verification Demonstration Test - The contractor shall conduct a coordinated maintainability testing program in accordance with requirements as defined in 4.3.3 of MIL-STD-471.

4.6.14.2.2 Maintainability Verification Test Plan - The contractor shall conduct a Maintainability Verification Test at the contractor's facilities in accordance with paragraph 4.3.3 of MIL-STD-471. The contractor shall prepare a Maintainability Verification Test Plan in accordance with 4.2 of MIL-STD-471 prior to the start of Maintainability Verification Test.

4.6.14.2.3 Maintainability Demonstration Test Plan - The contractor shall prepare a Maintainability Demonstration Test Plan in accordance with 4.2 of MIL-STD-471, prior to scheduled start of the Maintainability Demonstration Test. Navy Maintenance personnel shall be used for the test. Personnel skill levels will be established by the procuring activity.

4.6.14.2.4 Maintainability Test Procedures Report - The contractor shall prepare a Maintainability Test Procedures Report in accordance with MIL-STD-471, prior to starting the test.

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4.6.14.2.5 Maintenance Task Selection - The contractor shall prepare a list of realistic and representative maintenance tasks to be used in the Maintainability Demonstration Test in accordance with MIL-STD-471, Appendix A, and included in Maintainability Demonstration Test Plan.

4.6.14.2.6 Maintainability Test Report - The contractor shall provide a final demonstration report in accordance with MIL-STD-471.

4.6.15 Production tests - The Contractor shall perform all tests listed under Section 4 on not less than two (2) units out of each 50 units produced. The Government reserves the right to make the random selection. The Contractor can test four units concurrently when he so desires. Each production unit shall be tested in accordance with the requirements outlined in paragraphs 4.6.5, 4.6.6, 4.6.7.1, and 4.6.11.3. In addition, each test tractor shall be required to come to a complete stop from a speed of 10 mph within 20 feet. During these tests, all gauges, instruments, lights, horn and other operating components shall be tested to determine that they perform the function for which they are intended.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging - Preservation and packaging shall conform to MIL-STD-281, level A (Mobile) or level C (Mobile) as specified. The dash panel shall be adequately protected against transportation damage.

5.1.1 Level A - The tractor shall be cleaned and preserved in accordance with MIL-P-116.

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5.1.2 Level C - The tractor shall be preserved and packaged in accordance with the supplier's standard practice.

5.1.3 Packaging - Packaging shall comply with Consolidated Freight Classification Rules or other carrier regulations applicable to the mode of transportation.

5.2 Marking - In addition to any special marking required in the contract or order (see 6.2) marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use - Tractors covered by this specification are intended for use in towing aircraft and ground support equipment on concrete and asphalt surfaces at military airfields in all types of weather from -25° to +125°F.

6.2 Ordering data - Procurement documents should specify:

6.2.1 Procurement requirements -

(a) Title, number, and date of this specification.

(b) Level of preservation and packaging required (see 5.1).

(c) Special marking, if required.

(d) Contract data requirements, if required (see 6.2.3).

(e) First article inspection, if other than specified (see 6.2.2.1).

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(f) Where first article inspection shall be conducted (see 6.2.2.1(a)).

(g) Where first article inspection report shall be sent.

6.2.2 First article inspection - Contracts or orders should specify the following provisions for first article inspection.

6.2.2.1 Whether first article inspection is required - When a contractor is in continuous production of the tractor from contract to contract, consideration should be given to waive the first article inspections. If first article inspection is required, indicate:

(a) Where the first article inspection is to be conducted (at the contractor's plant, Government, or commercial facilities).

(b) That the approval of first article samples, or the waiving of the first article inspection, shall not relieve the contractor of his obligation to fulfill all other requirements of the specification and contract.

6.2.3 Contract data requirements - Items of deliverable data required by this specification are cited in the following paragraphs:

Paragraph	Data Requirement (Short Title)	Applicable DID
3.5.1.2	Reliability Program Plan	DI-R-2113
3.5.1.4	Reliability Test Plan	DI-R-2124
3.5.1.5	Reliability Test Result Report	DI-R-2118

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Paragraph	Data Requirement (Short Title)	Applicable DID
3.6.3	Maintainability Program Plan	DI-R-1740
3.6.5	Maintainability Status Reports	DI-R-1741

Such data will be delivered as described on approved (numbered) DID's (Data Item Description/DD Form 1664) only when specified on DD Form 1423 (Contract Data Requirements List) and incorporated into the applicable contract.

6.3 Definitions -

6.3.1 First article inspection - First article inspection consists of examinations and tests performed on the first tractor manufactured (which is representative of the production units) after the award of a contract. The examinations and tests are conducted to determine that the tractor meets the requirements of this specification (see 4.3).

6.3.2 Quality conformance inspection - Quality conformance inspection consists of examinations and tests performed on the production units to determine that the units meet the requirements of this specification (see 4.4).

Preparing Activity

Navy - AS

(Project No. 2420-N031)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, *government activities, or manufacturers/* vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-T-81038D TRACTOR, AIRCRAFT TOWING GASOLINE 7500 POUNDS DRAWBAR PULL,

NAME OF ORGANIZATION AND ADDRESS Type TA-75

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED S/N 0102-014-1802

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